



Using an MR-Based Teacher Orchestration Tool in AI-Supported K-12 Classrooms

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Abstract. Intelligent tutoring systems (ITSs) can personalize learning in K-12 classrooms, while teachers still need to make real-time decisions about how to allocate support across students. Mixed Reality (MR) teacher tools have shown promise for supporting this work, yet evidence from authentic classrooms remains limited. We report an in-the-wild field study of Lumilo2, an MR-based teacher orchestration tool integrated with MATHia and deployed in middle school mathematics classrooms. Across seven classroom sessions involving six teachers and 113 students, we examined how the system was used in practice and perceived by teachers and students. Teachers valued the tool's rapid, hands-free support for noticing student needs, while also wanting more specific, prioritized, and configurable information. Students were generally positive about their teacher's use of the tool, although some raised privacy concerns.

Keywords: Mixed Reality · Classroom · Teacher Orchestration Tool

1 Introduction

When K-12 students use intelligent tutoring systems (ITSs) in class, teachers must constantly decide whom to help and when to step in. This can be difficult because students' help requests do not always match their actual needs for support [2]. Analytics-based teacher tools such as wearable displays [9] and dashboards [15] can help by surfacing students' needs in real time. Prior work suggests that Mixed Reality (MR) tools [9] are promising for supporting this kind of in-the-moment orchestration. However, examining their use beyond one-off feasibility studies for scaling remains difficult because the orchestration demands that arise in one context often fail to transfer to another: fine-grained student logs available in one ITS may be absent in another [13]; interaction modalities that feel natural in lab studies may prove cumbersome in real classrooms [1];

and practices that are socially acceptable in one grade level may create stigma in another [14].

This paper addresses this gap by investigating how an MR-based orchestration tool can be adapted and appropriated when moved across tutoring platforms and classroom contexts. We implemented Lumilo2 to support productive co-orchestration between the teacher and the ITS. Building on prior work on Lumilo [9], Lumilo2 extends its core idea of using MR displays to support real-time teacher awareness. As a step toward routine classroom use without on-site researcher support, Lumilo2 was designed to generalize across a wider range of content and to support easy switching across classes and classrooms. The contributions of our study include: (1) empirical evidence from an in-the-wild field study on how an MR-based orchestration tool is used and perceived in authentic practice from both teachers ($N = 6$) and students ($N = 113$), and (2) design implications that surface and address the challenges of preparing MR orchestration tools to scale across diverse ITS-supported classroom contexts.



Fig. 1. Illustrative views of Lumilo2 use in authentic classrooms. (a) A teacher intervenes directly with students during ongoing AI-supported practice. (b) A student engages with the MATHia, generating real-time learning data. (c) An MR view of the classroom presenting real-time indicators of student learning status.

2 MR-Based Teacher Orchestration System

We implemented Lumilo2 (Fig. 1), an MR smart glasses application designed to work with an ITS to aid teachers in orchestrating personalized class sessions in which students work with the tutor at their own pace. It receives a real-time stream of student activity data and transforms this into indicators and detailed analytics shown in the MR interface. Lumilo2 started as a partial replication of Lumilo [9], an MR teacher-augmentation system previously validated in real classrooms using Lynnette [12] (an ITS for algebra practice). We reproduced a subset of Lumilo’s essential features on a different ITS (MATHia) and then extended with teacher-driven feature additions in a lighter, phone-powered MR device (XREAL Air 2 Ultra). Next, we describe the system’s core components.

Supported Intelligent Tutoring System. Lumilo2 integrates with a more widely deployed ITS, *MATHia* (see Fig. 1b), a large-scale ITS used by 600,000+

students in grades 6–12 across 2,000+ schools. Each grade-level course comprises about 100 topical “workspaces,” within which students must show their work by completing multi-step problems. MATHia includes a broad range of problem types, such as detailed word problems and interactive graphing problems.

Bridge Middleware. Between MATHia and Lumilo2 sits the *Bridge*, a tutor-agnostic middleware layer that decouples Lumilo2 from any single ITS. The Bridge ingests raw class and transaction logs from each ITS’s learning management interface, maps them to a unified event schema, and stores the result in a dedicated time series database. Rule-based detector algorithms are executed within this layer; they capture different aspects of student learning behaviors, which are crucial for understanding how teachers respond to students [6]. The Bridge features detectors (see Fig. 1c) for *idle* (no interactions in the software for 2 min or more [9]), *system misuse* (rapid guessing, abusing hints, or gaming the system [3]), *struggle* (students who have attempted the same skill three or more times), *critical struggle* (a stricter case in which students have attempted a skill three or more times without mastering it despite system support [4]) and *doing well* (low recent error rate in current activity, defined as an error rate less than 20% over the last 10 actions [9]). These detectors were based on open-source code [16] featured in Lumilo [9], which has been shown to significantly improve student learning gain when shown to teachers in classrooms where students learn with tutoring systems [9]. This middleware makes Lumilo2 extensible to additional ITSs and enables new features such as indicator prioritization and cloud-based classroom persistence.

MR-based Orchestration Tool. Lumilo2 enables five classroom-facing features: (1) Real-time Awareness: compact MR overlays above each student encode the current detector state in real time, allowing instructors to prioritize whom to help without approaching their desks. (2) Quick-dive Screens: teachers could access detailed quick-dive screens showing concrete information of student activity. (3) Classroom Persistence: at classroom setup time, the instructor places a single room-level spatial anchor that defines a global coordinate frame. All student indicators are registered relative to this anchor, so returning to the classroom on subsequent days automatically restores the virtual layout. (4) Interaction Methods. Teachers can switch between hand gestures for quick inspection and the phone controller for scrolling through historical detail or entering text. (5) Customization: options enabled teachers to prioritize indicators and notifications according to their preferences.

3 Methods

The study was approved by our university’s Institutional Review Board (IRB). This in-the-wild field study had two aims: (1) to place teachers in diverse classroom contexts (variation in grade level, curricular content, teaching configurations) to elicit rich, experientially grounded design feedback for future long-term scaling and (2) to investigate how students perceive their teacher’s use of MR glasses across these settings.

Participants and Settings. The participating site is located in a rural region of the eastern United States. A total of six middle school mathematics teachers (grades 5–8) participated in the study, with five different classrooms. Their teaching experience ranged from 2 to 16 years. A total of seven sessions were conducted, including two co-teaching sessions and two repeated sessions with the same teacher and class. During all sessions, the students used MATHia while the teacher (and their assistant, if there was one) used Lumilo2. LiveLab (a desktop-based dashboard for MATHia classes) is also available for teachers who choose to use it. 113 different students completed a post-class survey, of whom 48 participated in two sessions, with 44 students filling out the survey for both sessions. All participating teachers and students had prior experience using MATHia in their mathematics classrooms before the introduction of Lumilo2.

Study Procedure. Each class session lasted 45 min. A training session to help familiarize teachers with Lumilo2 was conducted separately from class time (one day in advance). During the lessons, students spent 35 min working individually on MATHia with Lumilo2 active in the classroom. Teachers supported their students while wearing Lumilo2, responding to real-time indicators, having one-on-one interactions with students they deemed in need of help, and generally engaging in classroom orchestration. Members of the research team conducted informal observations in the classroom to capture contextual information about teacher practices and student reactions. During the final 10 min of each class, students completed an online post-class survey (see Fig. 2). The survey also included open-ended questions (e.g., “How did you feel about your teacher using the glasses/technology?”). For students who attended two sessions, we added one extra open question to let them compare their experiences between the two sessions. We conducted individual semi-structured interviews with teachers (around 60 min) covering: (1) classroom context and how Lumilo-MATHia was used to support instructional goals; and (2) challenges, constraints, and considerations for using MR-supported orchestration tool in practice.

Data Collection and Analysis. Data came from three sources: (1) student post-class surveys, (2) semi-structured interviews with teachers, and (3) classroom observation notes. For student surveys, quantitative data were analyzed using descriptive statistics to summarize overall patterns. We used directed content analysis of responses to open-ended questions: two researchers independently mapped each comment to the survey constructs, adding an emergent tag when needed. For interviews, we applied reflexive thematic analysis [5]. We first familiarized ourselves with the dataset, then generated initial codes through open coding. Codes were iteratively clustered into candidate themes, which we reviewed and refined against the raw data for coherence and distinction. Finally, we defined and named the resulting themes, selected representative excerpts to illustrate each, and presented them below. Classroom field notes were coded alongside the teacher interview data and used to triangulate the themes emerging from the interviews.

4 Results

4.1 Teachers' Practices and Desires

Teachers reported that heads-up, spatially aligned cues helped them maintain awareness of the whole class while keeping their eyes on students, consistent with prior research (e.g., [11]). They also found that adding a positive indicator was useful for recognizing students who were doing well and encouraging them. The lighter-weight, phone-powered MR device clearly improved comfort; all teachers reported that wearing the XREAL glasses for a full lesson felt acceptable and that long-term use would likely be manageable. However, because the XREAL lenses still block some ambient light, we observed teachers tilt the glasses up occasionally, when providing one-on-one help and needing to see a student's screen directly. Besides these experiences, we identified the following themes in teachers' desires for the future design.

Desires for Flexible Teacher-Student Communication Channels. MR support reshaped how teachers managed visibility in teacher-student interactions. Consistent with prior work showing that visible help-seeking can create stigma or reduce participation [10,11], teachers in our interviews wanted students to be able to request help privately, without drawing attention from peers. They worried that public displays, such as projecting student status on a classroom screen, could embarrass students. As T1 noted, "6th grade... get a lot more embarrassed, so I didn't use it as much." At the same time, for positive indicators, teachers were enthusiastic about making them visible. In T3's class, whenever Lumilo flagged a student as doing well, T3 publicly noted their name and praised them. As T3 explained in the interview: "For these ['doing well' indicators], I want students to see them. It encourages those who don't have a 'doing well' signal yet to try to perform better so they can see theirs too."

Desires for Concrete and Rapidly Actionable Information About Student Difficulties. Consistent with prior work highlighting the limitations of abstract indicators in real-time classroom orchestration (e.g., [8,10]), teachers in our study expressed a strong desire for information that was concrete and immediately actionable. Notifications in the MR glasses did help surface struggling students more quickly than walking back to a LiveLab dashboard, yet the undifferentiated design of indicators often left teachers uncertain about whom to prioritize. This pushed teachers to rely more heavily on contextual knowledge about students. As T4 recalled, interpreting identical alerts in the first minutes of class felt "a little overwhelming, with three or four people that all needed help." Teachers thus asked for graded diagnostic indicators that distinguish the nature and severity of difficulties. T5 wished for thresholds that could be adjusted to highlight "the most important indicator for me to get to that kid before they sit there for 10 min." Teachers also wanted more nuanced, teacher-configurable support, such as adjustable thresholds and prioritization based on student history. As T1 noted, it would help to distinguish between "this kid is on Workspace 107 with a question mark" and "another kid still on 7 with a question mark."

Desires for Shared, Flexible Coordination in Co-teaching. In co-teaching classrooms, MR support changed coordination from fixed spatial division to more flexible, need-based coverage. Without the glasses, teachers often split the room by area: “you stay on that side, I’ll stay on this side, or you do the back row, I do the front row” (T1). However, when both wore the glasses, teachers reported that the immediacy of heads-up indicators encouraged shorter, more frequent check-ins with students. T1 and T4 both described moments when they changed direction mid-walk because a new indicator appeared. This micro-orchestration pattern did not occur when they used LiveLab, which required looking away from students. As T1 noted, with Lumilo, “Because we both could see who was struggling, we weren’t stuck on just our half. If I saw a kid in the back having trouble, I could go, even if it was usually her side. She could do the same. We could adjust based on need instead of location.” In co-teaching classrooms where only one teacher wore the glasses, the tool still mediated collaboration, but asymmetrically. T3 explained, “My assistant didn’t have the glasses, but I could see right away who needed help and tell her, ‘Go check on so-and-so.’ ” This created what T3 described as a complementary partnership: “I could scan the class, and she could stay with a student longer, because I was letting her know who else needed it.”

4.2 Student Attitudes Toward Teacher Use of MR Glasses

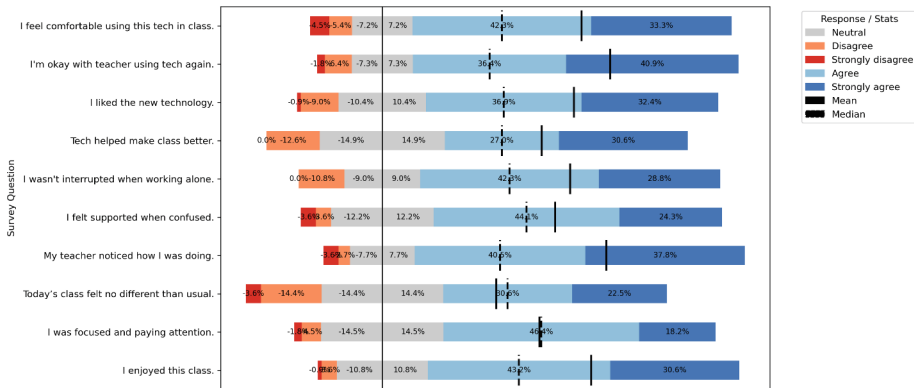


Fig. 2. First-session post-class student survey (repeated participants excluded; N = 113). Stacked bars show response distributions on a 5-point Likert scale.

Survey results (Fig. 2) indicated that students were generally positive about their teacher’s use of MR glasses in class. For comfort and acceptance, about three-quarters of students agreed or strongly agreed that they felt comfortable using the technology in class (76%) and were okay with their teacher using it again (77%), and around 70% liked the new technology. For perceived instructional benefit, a majority felt that the technology helped make class better (58%)

and that they were not interrupted when working alone (71%); roughly two-thirds agreed that they felt supported when confused (68%). Regarding perceived teacher support, about 78% agreed or strongly agreed that their teacher noticed how they were doing. For attention, around 65% reported that they were focused and paying attention. For overall enjoyment and normalcy, roughly 74% said they enjoyed the class, while responses to “Today’s class felt no different than usual” were more mixed: over half agreed or strongly agreed (53%). The pattern suggests that most students were comfortable with the MR setup, noticed concrete benefits especially feeling seen and supported, and did not find the class very disruptive.

Open-ended responses provided further nuance. Many students described the technology as engaging and supportive: “I liked when Miss [T1] came to help me when I was stuck on a question...she saw I was struggling and came to help me out.” Students also highlighted the emotional dimension of feeling seen, with one remarking, “I liked seeing me having smiley faces and doing good because it made me feel good and want to work even harder.” Several comments reflected the novelty and excitement of seeing their teacher use MR glasses: “Funny to see Mrs. [T2] in goofy glasses,” and “It made me feel more interested on how the glasses functioned.” At the same time, some students expressed concerns about privacy or feeling watched: “I felt uncomfortable during it and felt like someone was going to be called out on something that should be kept private.” A few students worried about cameras in the room (misinterpreted the glasses as recording): “The only thing that felt different...was there is a camera in the room. I would change nothing except removing the cameras.” When asked if they felt more noticed, most students said yes, often connecting this feeling directly to the MR glasses: “After the teacher used the special glasses, I felt more noticed,” and “I did not need to raise my hand—she came to me when I was struggling.” However, a small group reported feeling less noticed. For example: “I felt the same as always, but less [noticed] because he had to help the others.”

Among students who participated in two sessions and filled out the survey after each ($N = 44$), we conducted a within-subject comparison of survey responses. Using a Wilcoxon signed-rank test, we found no significant differences between two sessions across all survey items. Analysis of open-ended responses further reinforced this finding. Most students reported that the second session “felt the same” as the first, with many stating there was “nothing different” about today’s class. A smaller set of students highlighted minor differences that the second session felt more “normal” or that they had become “used to” the glasses.

5 Discussion

Our results show that teachers want customized, actionable support that helps them respond quickly when many students need attention. This echoes prior work on teaching augmentation [2] and suggests the following directions. First,

make suggestions glanceable and prioritized. For example, a compact micro-card could summarize the trigger and brief evidence, time since onset, and a recommended next step (e.g., encouragement, check-in, step-by-step help). Second, elevate small-group opportunities as first-class actions: when multiple students show similar difficulties, the system can recommend forming a mini-lesson group and provide prompts plus a simple co-teaching handoff (e.g., [17]). Third, provide customizable prioritization tools that let teachers tune thresholds and indicator definitions (e.g., by adjusting the time window used to define idle). Future systems could also use AI to recommend which students most need attention, drawing on teacher feedback, prior triage decisions, and other contextual inputs.

Our results show that teachers are highly sensitive to the social visibility of analytics in class: they wanted students to be able to request help discreetly and worried that public displays could embarrass students. Students also have concerns about privacy and being watched, consistent with prior work on AI-teacher collaboration [7, 10, 11]. Future designs should therefore make visibility rules explicit and teacher-configurable: teachers should be able to decide who sees what information, and when. MR orchestration tools should keep individual-level analytics private by default on the teacher’s device and support other reveals, such as a private prompt to the teacher and optional disclosure (aggregate or per-student) when the teacher opts in, aligning with calls for selectively shared analytics [8].

Currently, Lumilo2 provides a single teacher-facing view. For broader classroom adoption, future versions could add teacher-configurable role views: a co-teacher MR view with a synchronized triage list and shared handoff, a student view in the laptop that supports discreet help-seeking without public exposure, and an administrator view that reports high-level usage summaries rather than evaluative traces. In addition, to support a bidirectional, low-friction backchannel, future systems could also offer MR-based teacher-student signaling: “invisible hand raise” from the student device; one-tap private acknowledgment from the teacher; lightweight “not now/come back soon” replies; and silent haptics or tiny on-screen badges to avoid public stigma.

6 Conclusion and Future Work

We presented Lumilo2, an MR-based teacher orchestration tool for AI-supported classrooms, and reported findings from an in-the-wild deployment in middle school mathematics classrooms. Teachers valued rapid, heads-up awareness of student needs, but also wanted more glanceable and prioritized support, flexible control over the visibility of students’ status, and better support for co-teaching. Students were generally positive, although some raised privacy concerns. Future work could extend this study through longer, multi-site deployments and broader stakeholder involvement, such as administrators and parents, to examine their views on the adoption of MR-based orchestration tools.

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